

Chapter 2: Essential Algebra & Equations

Polynomial Manipulation, Quadratic Equations, Systems of Equations & Factoring Identities

2.1 Essential Algebraic Identities & Factoring

Fluency with standard algebraic identities allows rapid simplification of rational expressions and polynomial manipulation.

Quadratic Identities

- *Difference of Squares:* $a^2 - b^2 = (a - b)(a + b)$
- *Perfect Square:* $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- *Sum of Squares:* $a^2 + b^2 = (a + bi)(a - bi)$

Cubic Identities

- *Difference of Cubes:* $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- *Sum of Cubes:* $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- *Cube of Binomial:* $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$

2.2 Quadratic Equations & The Discriminant

For any quadratic equation in standard form $ax^2 + bx + c = 0$ (where $a \neq 0$), solutions are given by the Quadratic Formula.

Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Discriminant (Δ): $\Delta = b^2 - 4ac$

Nature of Roots Based on Discriminant (Δ)

- $\Delta > 0$: Two distinct real roots.
- $\Delta = 0$: One repeated real root ($x = -b / 2a$).
- $\Delta < 0$: Two complex conjugate roots.

Worked Example 2.1: Quadratic Analysis

Problem: Determine the nature of the roots and solve: $2x^2 - 4x + 5 = 0$.

Solution: $a = 2$, $b = -4$, $c = 5$.

$\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$ ($\Delta < 0 \Rightarrow 2$ complex roots).

$x = \frac{4 \pm \sqrt{-24}}{4} = \frac{4 \pm 2i\sqrt{6}}{4} = 1 \pm (i\sqrt{6})/2$.

2.3 Systems of Linear Equations

Systems of linear equations can be solved using Substitution, Elimination, or Matrix methods (Cramer's Rule).

Worked Example 2.2: Elimination Method

Problem: Solve the system:

(1) $3x + 2y = 11$

(2) $5x - y = 14$

Solution: Multiply equation (2) by 2:

$$10x - 2y = 28.$$

Add to equation (1): $(3x + 10x) + (2y - 2y) = 11 + 28 \Rightarrow 13x = 39 \Rightarrow \mathbf{x = 3}$.

Substitute $x = 3$ into (2): $5(3) - y = 14 \Rightarrow 15 - y = 14 \Rightarrow \mathbf{y = 1}$.

Solution set: **(3, 1)**.

2.4 Vieta's Formulas & Polynomial Roots

Vieta's formulas establish relations between the coefficients of a polynomial and its roots.

Quadratic Polynomial Relations

For $ax^2 + bx + c = 0$ with roots r_1, r_2 :

- Sum of roots: $r_1 + r_2 = -b/a$
- Product of roots: $r_1 \cdot r_2 = c/a$

Chapter 2 Review & Practice Problems

1. Factor completely over the real numbers: $x^4 - 16$ and $8x^3 + 27$.
2. Find all real and complex solutions to $x^2 + 6x + 13 = 0$ by completing the square.
3. Solve the 2x2 linear system:
$$2x - 3y = 7$$
$$4x + 5y = 3$$
4. If the roots of $3x^2 - kx + 12 = 0$ are equal, find the possible values of k .
5. Simplify the rational expression: $(x^2 - 9) / (x^2 - 5x + 6)$.

